

Jiahao Luo

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3D/4D Representation Learning Researcher | State Estimation, Embodied Dynamics, and World Modeling

RESEARCH SUMMARY

Ph.D. candidate in computer vision, graphics, and multimodal generative AI, studying state representations for dynamic 3D/4D systems. My research focuses on learning compact representations that factor persistent structure from dynamic motion, estimating these states from multimodal and partial observations, and using them for controllable generation, reconstruction, and real-time simulation. I develop transformer and diffusion models for 3D/4D reconstruction, motion modeling, representation learning, and generative animation, together with scalable synthetic-data pipelines that expand supervision and real-world generalization. My long-term interest is in extending these representations from individual dynamic entities toward action-conditioned human-centric world models.

EDUCATION

University of California, Santa Cruz - Santa Cruz, CA, USA

Ph.D. in Computer Science and Engineering, advised by Prof. James Davis | Sep 2021 - Expected Jan 2027

University of California, Santa Cruz - Santa Cruz, CA, USA

M.S. in Electrical and Computer Engineering | Sep 2018 - Jun 2020

Beijing University of Posts and Telecommunications (BUPT) - Beijing, China

B.Eng. in Electronic Engineering | Sep 2014 - Sep 2018

RESEARCH EXPERIENCE

ByteDance Inc. - Research Intern, San Jose, CA | Jun 2026 - Present

Mentor: Chengjiang Long

- Developed learned state representations for high-fidelity dynamic humans, replacing 3DMM parameterizations with expressive latent states that capture fine-grained, nonlinear facial dynamics while separating persistent identity from time-varying behavior.
- Established two paradigms for constructing human state from observations: Image2Latent infers a template-free dynamic state directly from multi-view RGB, while GS2Latent encodes 3D Gaussian deformation into a compact 256D latent representation.
- Developed a real-time state observer that recovers the same learned dynamic representation from sparse first-person infrared views, enabling 3D human-state estimation from highly partial online observations.
- Extended a feed-forward multi-view reconstruction model for structured 3D human-state estimation, and improved its in-the-wild generalization through a scalable synthetic-data engine that post-trained a video generation model to transform unconstrained portraits into 3D-consistent multi-view static and dynamic sequences.

Snap Inc. - Research Intern, Santa Monica, CA | Sep 2024 - Jun 2026

Mentors: Peter Wonka, Jian Wang, Hsin-Ying Lee

- Led research on structured state representations and controllable dynamics for deformable 3D entities, spanning cross-embodiment state spaces, generative motion supervision, 4D reconstruction, and real-time simulation.
- Built a large-scale cross-embodiment state dataset spanning 33,724 non-humanoid identities and 10 semantically aligned animation states per identity, using generative image and 4D pipelines to capture how shared behavioral states are realized across highly diverse geometries and appearances.
- Developed RegHead, a feed-forward registration framework that learns a shared semantic state and control space from unregistered data, using stochastic-anchor representations and local/global correspondence to align heterogeneous geometry without requiring common topology or a predefined skeleton.
- Enabled real-time behavior transfer from human performance to diverse embodiments by mapping tracked human motion into the shared control space and decoding it into embodiment-specific dynamic states.
- Developed T2Bs to convert static generated 3D assets into controllable dynamic 4D representations, using text-conditioned video generation and view-conditioned deformable Gaussian reconstruction to transform generated motion observations into temporally coherent 4D state trajectories.
- Contributed to RigMo, learning structure-aware rig and motion latent spaces that factor persistent articulation structure from time-varying motion, together with Motion-DiT for generative modeling directly in the learned motion state space.
- Contributed to feed-forward multi-view 3D reconstruction and Gaussian-domain adaptation for real-time stylized 3D generation, and transferred the controllable 4D generation pipeline into Snap's Lens Studio production workflow.

Flawless AI - Research Intern, Santa Monica, CA | Jun 2024 - Sep 2024

Mentor: Kyle Olszewski

- Developed 4D Gaussian representations that factor persistent intrinsic appearance and material properties from time-varying illumination state, enabling dynamic 3D reconstruction with independently controllable scene appearance.
- Built a synthetic illumination data engine that observes diverse environment lighting conditions, providing controlled supervision for learning disentangled geometry, material, and illumination representations.

Bellus3D - Software Engineer, Campbell, CA | Oct 2020 - Aug 2021

Mentor: Eric Chen

- Built a large-scale mobile 3D dataset and transformed heterogeneous reconstructed geometry into a canonical UV-space representation, providing a structured state space for dense image-to-geometry learning.
- Developed an RGB-to-3D regression model that recovers structured facial geometry from a single visual observation and integrated it into a production mobile reconstruction pipeline.

SKILLS

Representation learning and dynamic modeling: Latent state representations, multimodal state estimation, structure-motion factorization, representation disentanglement, motion modeling and generation, correspondence learning, controllable deformation, feed-forward prediction, and temporal modeling

3D/4D vision and generative modeling: 3D/4D reconstruction and generation, dynamic scene representations, Gaussian splatting, neural rendering, transformers, diffusion and flow models, video generation, multi-view learning, mesh and point-cloud processing, and feed-forward registration

Data-centric learning and evaluation: Synthetic data generation, dataset construction and curation, multimodal alignment, weak/pseudo supervision, data-quality validation, semantic annotation, controlled experiments, ablation studies, automated evaluation, and human evaluation

Research engineering and deployment: Python, PyTorch, distributed and multi-GPU training, scalable inference pipelines, experiment tracking, visualization, reproducible research workflows, CUDA-aware deployment optimization, and research-to-product integration

AWARDS

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| • WACV 2025 Oral presentation | Feb 2025 |
| • CVPR 2022 Oral presentation | Jun 2022 |
| • UCSC Chancellor's fellowship | Mar 2021 |

SELECTED PUBLICATIONS

- **RigMo: Unifying Rig and Motion Learning for Generative Animation**
[CVPR 2026](#)
Hao Zhang, **Jiahao Luo**, Bohui Wan, Yizhou Zhao, Zongrui Li, Michael Vasilkovsky, Chaoyang Wang, Jian Wang, Narendra Ahuja, Bing Zhou
- **RegHead: Non-Humanoid Head Blendshapes via Feed-Forward Registration**
[ECCV 2026](#)
Jiahao Luo, Hao Zhang, Jianqi Chen, Yijie He, Jiaxu Zou, Michael Vasilkovsky, Sergei Korolev, Sergey Tulyakov, Chaoyang Wang, Peter Wonka, James Davis, Jian Wang
- **T2Bs: Text-to-Character Blendshapes via Video Generation**
[ICCV 2025](#)
Jiahao Luo, Chaoyang Wang, Michael Vasilkovsky, Vladislav Shakhrai, Di Liu, Peiye Zhuang, Sergey Tulyakov, Peter Wonka, Hsin-Ying Lee, James Davis, Jian Wang
- **FFAvatar: Few-Shot, Feedforward, and Generalizable Avatar Reconstruction**
Thuan Hoang Nguyen, **Jiahao Luo**, Yinyu Nie, Haoyang Wang, George Papandreou, Hao Li, Jian Wang, Gordon Guocheng Qian
- **Snapmoji: Instant Generation of Animatable Dual-Styled Avatars**
[WACV 2026](#)
Eric Ming Chen, Di Liu, Sizhuo Ma, Michael Vasilkovsky, Bing Zhou, Qiang Gao, Wenzhou Wang, **Jiahao Luo**, Dimitris N. Metaxas, Vincent Sitzmann, Jian Wang
- **SplatFace: Gaussian Splat Face Reconstruction Leveraging an Optimizable Surface**
[WACV 2025 Oral](#)
Jiahao Luo, Jing Liu, James Davis
- **Disjoint Pose and Shape for 3D Face Reconstruction**
[ICCV workshop 2023](#)
Raja Kumar*, **Jiahao Luo***, Alex Pang, James Davis
- **How much does input data type impact final face model accuracy?**
[CVPR 2022 Oral](#)
Jiahao Luo, Fahim Hasan Khan, Issei Mori, Akila de Silva, Eric Sandoval Ruezga, Minghao Liu, Alex Pang, James Davis.
- **How accurate is 2-view stereo to reconstruct a 3D face model?**
[ICIP 2022](#)
Jiahao Luo, Eric Ruezga, James Davis
- **DuelGAN: A Duel Between Two Discriminators Stabilizes the GAN Training**
[ECCV 2022](#)
Jiaheng Wei, Minghao Liu, **Jiahao Luo**, Andrew Zhu, James Davis, Yang Liu.
- **Low-light Image Enhancement Using Chain-consistent Adversarial Networks**
[ICPR 2022](#)
Minghao Liu, **Jiahao Luo**, Xiaohan Zhang, James Davis, Yang Liu
- **Face Models: How good does my data need to be?**
[ICIP 2021](#)
Jiahao Luo, Fahim Khan, Issei Mori, Akila de Silva, Eric Ruezga, James Davis.